



The constraint is linear: we must deliver dose 1 via $x_1 \cdot a_1 + x_2 \cdot a_2 = 1$, with $a_1 = 1$. Here the code makes $a_2 = 2f(\infty) + \frac{1}{2}$, and one checks by the geometric-series formula that

$$f(\infty) = 72^{-10^{24}} + \sum_{k=3}^{\infty} (1/2)^k - 72^{-10^{24}k} \approx 0.25$$

(since $\sum_{k=3}^{\infty} (1/2)^k = (1/2)^3 / (1 - 1/2) = 0.25$ ¹, and the $72^{-10^{24}k}$ terms are astronomically tiny). Thus $a_2 \approx 2 \cdot 0.25 + 0.5 = 1.0$ up to a negligible error. In fact $a_2 = 1 + \delta$ with $\delta > 0$ extremely small.

For a linear program “minimize $x_1 + x_2$ subject to $x_1 + (1 + \delta)x_2 = 1$, $x_i \geq 0$ ”, the faster rate is $a_2 > 1$, so the optimum uses only the second therapy (set $x_1 = 0$). In that case $x_2^* = 1/a_2 \approx 1$. (Indeed in the textbook example “min $x_1 + x_2$ ” with a similar linear constraint, the optimal solution has $x_1 = 0$ ².)

Thus an optimal point is $(x_1^*, x_2^*) = (0, 1/a_2) \approx (0, 1)$. The infinity-norm distance from $(0.1, 0.9)$ to $(0, 1/a_2)$ is

$$\max(|0.1 - 0|, |0.9 - 1/a_2|) \approx \max(0.1, |0.9 - 1|) = 0.1.$$

Since $0.1 < 0.15$, the given point $(0.1, 0.9)$ is indeed within 0.15 (in the infinity norm) of a true minimiser.

Answer: Yes. The optimal solution is essentially $(0, 1)$, so $\|(0.1, 0.9) - (0, 1)\|_{\infty} = 0.1 < 0.15$. ¹ ²

¹ Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series

² math.purdue.edu

<https://www.math.purdue.edu/~yipn/421/BECK.BECK.FIRST.COURS.LIN.OPT-Ch1-LinProgForm.pdf>